

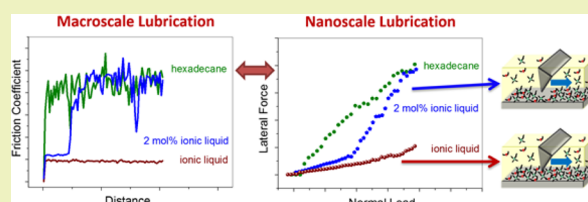
Combined Nano- and Macrotribology Studies of Titania Lubrication Using the Oil-Ionic Liquid Mixtures

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S Supporting Information

ABSTRACT: The lubrication of titania surfaces using a series of ionic liquid (IL)–hexadecane mixtures has been probed using nanoscale atomic force microscopy (AFM) and macroscale ball-on-disk tribometer measurements. The IL investigated is trihexyl-(tetradecyl)phosphonium bis(2,4,4-trimethylpentyl)phosphinate, which is miscible with hexadecane in all proportions. At both length scales, the pure IL is a much more effective lubricant than pure hexadecane. At low loads, which are comparable to common industrial applications, the pure IL reduces the friction by 80% compared to pure hexadecane; while the IL–hexadecane mixtures lubricate the titania surface as effectively as the pure IL and wear decreases with increasing IL concentration. At high test loads the adsorbed ion boundary layer is displaced leading to surface contact and high friction, and wear is pronounced for all IL concentrations. Nonetheless, the IL performs better than a traditional zinc–dialkyl–dithiophosphate (ZDDP) antiwear additive at the same concentration.

KEYWORDS: Ionic liquid, Additives, Lubrication, Titania, ZDDP



■ INTRODUCTION

Titanium and its alloys have wide applications in industry due to their high strength-to-weight ratio and high melting point.^{1,2} An oxidized titania layer is typically present on the titanium surface which prevents the bulk titanium from further corroding and oxidizing.³ However, the titania surface on top of the bulk titanium is prone to high friction and wear, and currently there are no suitable lubricants for titanium surfaces,^{4,5} which restricts their widespread use in moving parts.^{6,7} Conventional mineral oil based lubricants adhere weakly to titania, so they are easily squeezed out during sliding, leading to wear.⁸ Zinc–dialkyl–dithiophosphates (ZDDP) are commonly used antiwear lubricant additives for cast iron and steel, where they form a tribofilm. ZDDP does not form effective tribofilms on titania, and abrasive wear results.^{4,9}

Ionic liquids (ILs) are salts composed entirely of ions which are liquid at temperatures less than 100 °C.¹⁰ Compared with common molecular liquids and common lubricant base oils, IL ions adsorb strongly to a variety of solid surfaces via combinations of van der Waals, electrostatic, and H-bonding interactions.^{11,12} ILs often have sponge-like nanostructures in the bulk, which are flattened by the presence of a macroscopic flat surface.^{13–18} Solid–IL interfaces can be divided into three distinct regions: the boundary (surface-adsorbed) layer, the transition zone, and the bulk phase.¹⁵ The boundary (surface-adsorbed) layer consists of a single layer of ions in direct contact with the solid surface. The composition of this layer strongly

depends on the properties of the IL and the solid substrate. The transition zone consists of several layers through which the pronounced boundary layer structure decays to the bulk morphology. The boundary layers have significant effects on interparticle forces and related stability of colloids in ILs.^{19,20} Particle suspensions have found to be highly stable in ILs without any added stabilizer due to the overlap of the boundary layers formed on the interface.^{21,22}

ILs are promising lubricants due to their favorable physical properties. Because they have much higher affinity for solid surfaces than conventional mineral oil based lubricants,^{23–25} ILs form robust boundary layers that resist “squeeze-out” up to high normal load.²⁶ They have high mechanical and thermal stability, and excellent thermal conductivity, meaning they can transfer heat away from the sliding surfaces more effectively than conventional lubricants.²⁷ However, ILs are relatively expensive compared with typical oil based lubricants, and this limits their industrial use. The recent development of oil-miscible ILs provides the opportunity to use ILs as lubricant additives.^{28,29} Compared with commercially available additives such as ZDDP, ILs have no ash upon decomposition, better stability at high temperatures, and better compatibility with a wider range of surfaces.^{8,30–32}

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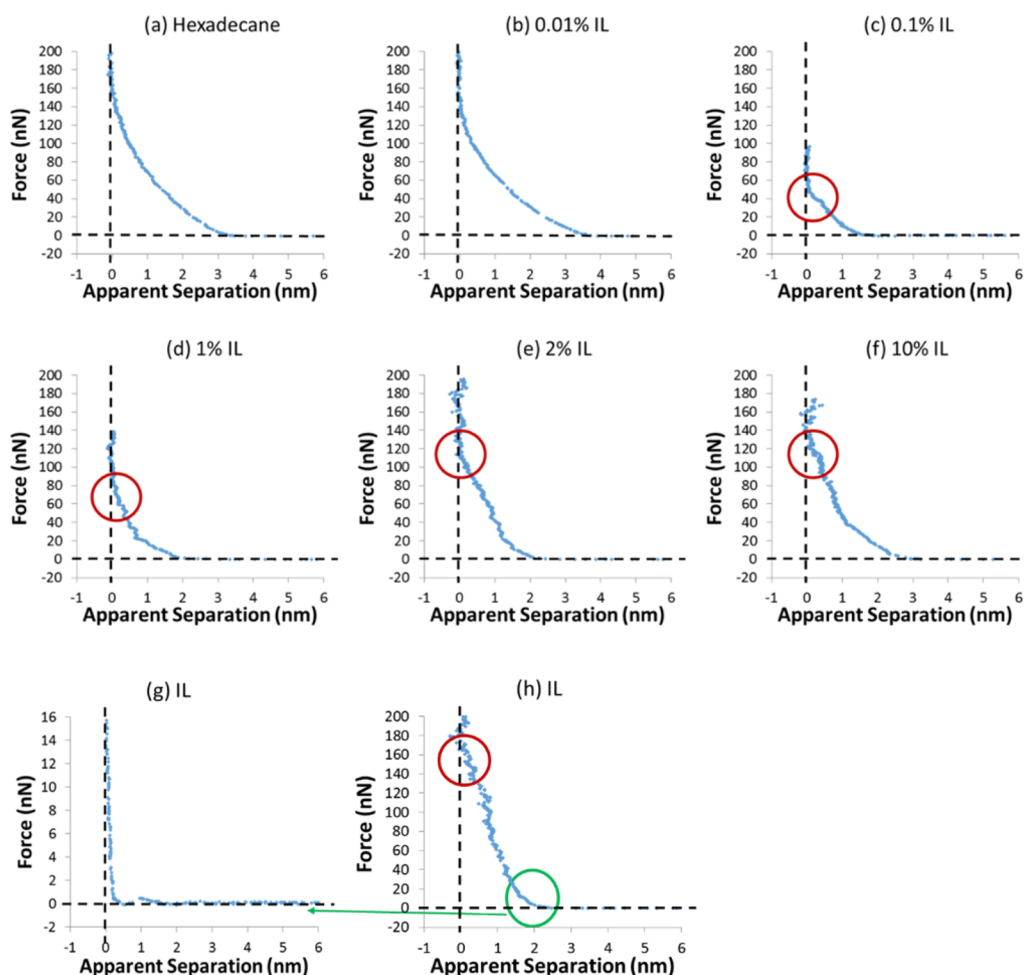


Figure 1. Typical normal force versus apparent separation profile for an AFM tip approaching a titania surface in pure hexadecane, pure IL, and IL–hexadecane mixtures.

Macrotribology studies have shown that IL–oil mixtures reduce friction and wear more effectively than neat base oils in several systems,^{8,28–30,32–35} but the mechanism by which IL additives lubricate a surface remains only partially understood.⁸ Nanotribology studies are less affected by surface asperities and roughness than macroscale measurements and, thus, can assist in the elucidation of lubrication mechanisms.³⁶ Recently we investigated both nano- and macrotribology of IL–oil mixtures at silica surfaces and revealed that the investigated IL (trihexyltetradecyl)phosphonium bis(2,4,4-trimethylpentyl)-phosphinate, $P_{6,6,6,14}(C_8)_2PO_2$ is an effective lubricant additive at both nano- and macroscale.^{37,38} This IL is chemically stable to both air and water and is fully miscible with apolar base oils.⁸

The lubrication of pure ILs on titania surfaces have been rarely investigated,^{3,39–41} and to date, the lubrication of titania using IL–oil mixtures has not been studied. It is not clear how IL ions interact with titania surfaces and lubricate the surface at either nano- or macroscales. In this work, we address this issue by investigating $P_{6,6,6,14}(C_8)_2PO_2$ as a lubricant additive for titania surfaces at both nano- and macroscales and compare its performance with the traditional lubricant additive, ZDDP. Hexadecane is used as a model apolar lubricant base oil, because of its simple structure, well-known properties, and it is readily available in high purity commercially. The outcomes underpin the development of new, cost-effective lubricant additives for titanium and alloy systems.

MATERIALS AND METHODS

Titanium alloys have wider applications than pure titanium metal due to higher strength.¹ A commercially available titanium alloy (AMS 4928 6AL/4 V, 90% Ti, 6% Al and 4% V, plus other trace elements) shows comparable friction with pure titanium in air; the difference in friction coefficients is within 10%.⁴² This suggests the oxidized layer on top of the titanium alloy is mainly titania, and the influence of Al and V components on friction is not significant. In this study, a titanium alloy (AMS 4928 6AL/4 V, supplied by Airport Metals, Australia) was chosen as the model surface in macrotribology tests, thus the obtained results are more relevant to industrial applications. The titanium alloy surfaces were prepared by polishing down to 2400grit grinding paper, resulting in an average surface roughness of 200 ± 50 nm by a profilometer. However, titanium alloy samples are not suitable for nanotribology tests due to difficulty in processing and high roughness, thus titania surfaces prepared by atomic layer deposition (ALD) on silicon wafers with titanium isopropoxide and water as precursors^{43,44} were used in the nanotribology tests for comparison. The thickness of the titania layer is ~ 70 nm. The morphology of the titania coated layer is shown in Figure S1 of the Supporting Information. The surface roughness is 1.1 ± 0.2 nm for a $250 \text{ nm} \times 250 \text{ nm}$ scan area. Hexadecane (ReagentPlus, 99%) was supplied by Sigma-Aldrich. The IL, $P_{6,6,6,14}(C_8)_2PO_2$ (purity > 95%) investigated in this study was supplied by Iolitec. ZDDP was supplied by Nanjing SAMWON International Limited, China (water content <0.04%).

Nanotribology measurements were performed using a Bruker NanoScope VIII Multimode AFM. Three sharp silicon cantilevers (spring constant = 2.0 ± 0.2 N/m by the thermal tune method, tip radius ~ 8 nm) from the same batch (model NSC36, Mikromasch,

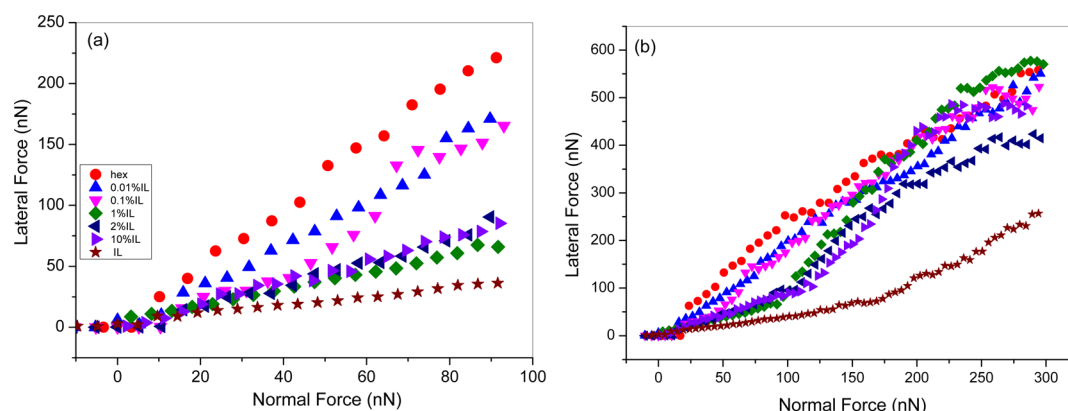


Figure 2. Lateral force versus normal load in IL–hexadecane mixtures for an AFM tip ($r \approx 8$ nm) sliding on a titania surface at $6 \mu\text{m s}^{-1}$. (a) Data up to 90 nN normal load. (b) Data up to 300 nN.

Tallinn, Estonia) were used over the course of the investigation. The cantilever was cleaned before use by Milli-Q water, dried under nitrogen, and irradiated with ultraviolet light for 20 min to remove organic contaminants. The sample volume injected in the AFM liquid cell was 0.1 mL. Normal force curves were collected by moving the surface toward the tip and detecting the cantilever deflection as a function of separation. The ramp sizes were 150 nm and ramp rate was 0.2 Hz. Standard methods were used to convert deflection vs separation data to normal force vs apparent separation curves.⁴⁵ Lateral forces were obtained in contact mode by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled.⁴⁵ The scan size was 100 nm, and scan rate was 30 Hz. The lateral deflection signal (i.e., cantilever twist) was converted to lateral force using a customized function produced in Matlab R2015a which takes into account the torsional spring constant and the geometrical dimensions of the cantilever.⁴⁶ These experiments were repeated more than three times; the obtained lateral force data showed the same trend with small variations.

Macrotribology was tested using a Nanovea (Irvine, CA) ball-on-disk tribometer using 3 mm radius Si_3N_4 balls on titanium alloys at room temperature. The sliding speed was 50 mm/s, and sliding distance was 20 m. The macroscale friction coefficient was recorded throughout the experiment. Experiments were conducted at loads of 2 and 10 N, respectively. A series of IL–hexadecane and ZDDP–hexadecane mixtures were added to the titanium alloy surfaces before the loaded Si_3N_4 ball was placed on its surface. On completion of the macrotribology tests the depths of the wear tracks were measured using a Bruker GT-K1 Optical Profiler and the morphologies of the wear tracks were measured using a NeoScope JCM-5000 scanning electron microscope (SEM).

RESULTS AND DISCUSSION

The interfacial nanostructure, especially the IL boundary layer of the IL–hexadecane mixtures, plays a key role in determining the lubricity.^{13,18,37,38} In this study we first investigated the interfacial nanostructure of IL–hexadecane mixtures by AFM normal force–separation profiles, then examined the nanotribology using AFM friction measurements and macrotribology using ball-on disk tests.

Interfacial Nanostructure. The interfacial nanostructure of pure hexadecane, pure IL, and IL–hexadecane mixtures at titania was probed by AFM normal force–separation profiles, which are presented in Figure 1.

For pure hexadecane, a zero constant force is detected beyond 3.5 nm, then the force increases monotonically as the AFM tip moves closer to the surface (Figure 1a). For the titania (surface)–hexadecane–silica (AFM tip) system investigated

here, as the Hamaker constants for hexadecane (5.2×10^{-20} J) and silica (6.5×10^{-20} J) are close in air,⁴⁷ the van der Waals component is very small;^{44,48–51} the origin of this short-range repulsive force is likely caused by repulsions between the surfaces.⁵² For all systems adhesive forces on retraction were weak, which is consistent with the friction data presented below.

Data for the pure IL is shown in Figure 1g and h. The data in Figure 1g was obtained using a small (30 nm) ramp size to reveal weak structural forces associated with interfacial IL nanostructure; similar experiments for all IL in oil mixtures produced featureless force curves. Figure 1h was obtained using a 150 nm ramp size, as per Figure 1a–f. In Figure 1g, one very weak step, with width equal to the ion pair cubic packing dimension (~ 1.2 nm), is detected and followed by a slight compression. This is similar to previous studies of pure ILs,^{15,16,26} but the number of steps in the force curve, the magnitude of the push through forces and the clarity of the steps are markedly reduced. This is because compared to conventional ILs this oil miscible IL is weakly associated and packs poorly into near surface structures.

For the pure IL when a large ramp size is used (Figure 1h), a push-through is noted at ~ 160 nN (indicated by the red circle) with a width of ~ 1.5 nm due to an additional ion pair layer being displaced. At first glance this push-through is somewhat unpersuasive, but the force at which it occurs is well matched with a breakpoint in the nanotribology data presented below, as are similar push-throughs for the various IL–oil mixtures. In previous studies of interfacial nanostructure we found that the AFM tip was unable to push through the final IL boundary layer and move into contact with the solid surface for a variety of surfaces, even at very high force.^{14,16,46,53} This is especially true for charged substrates due to strong binding of the IL ions to surface charged sites.^{15,46,54} Thus, generally the “zero” apparent separation corresponds to the AFM tip in contact with an impenetrable IL boundary layer, and the final detected step corresponds to the last push-through ion/ion pair layer.^{15,54} However, titania has a lower surface charge density than the previously investigated substrates like silica and mica.^{55,56} Thus, the density of IL ions electrostatically bound to titania is reduced, as are lateral van der Waals interactions between ions in the adsorbed layer. This means it is possible that the adsorbed ion layer on titania could be displaced at high force leading to direct tip–surface contact; the nanotribology results presented in Figure 2 confirm that this is the case.

Table 1. Nanoscale Friction Coefficients of IL–Hexadecane Mixtures at the Low Normal Load Regime^a

	hexadecane	0.01% IL	0.1% IL	1.0% IL	2.0% IL	10% IL	pure IL
μ	2.0	1.8	0.9	0.8	0.9	0.9	0.4

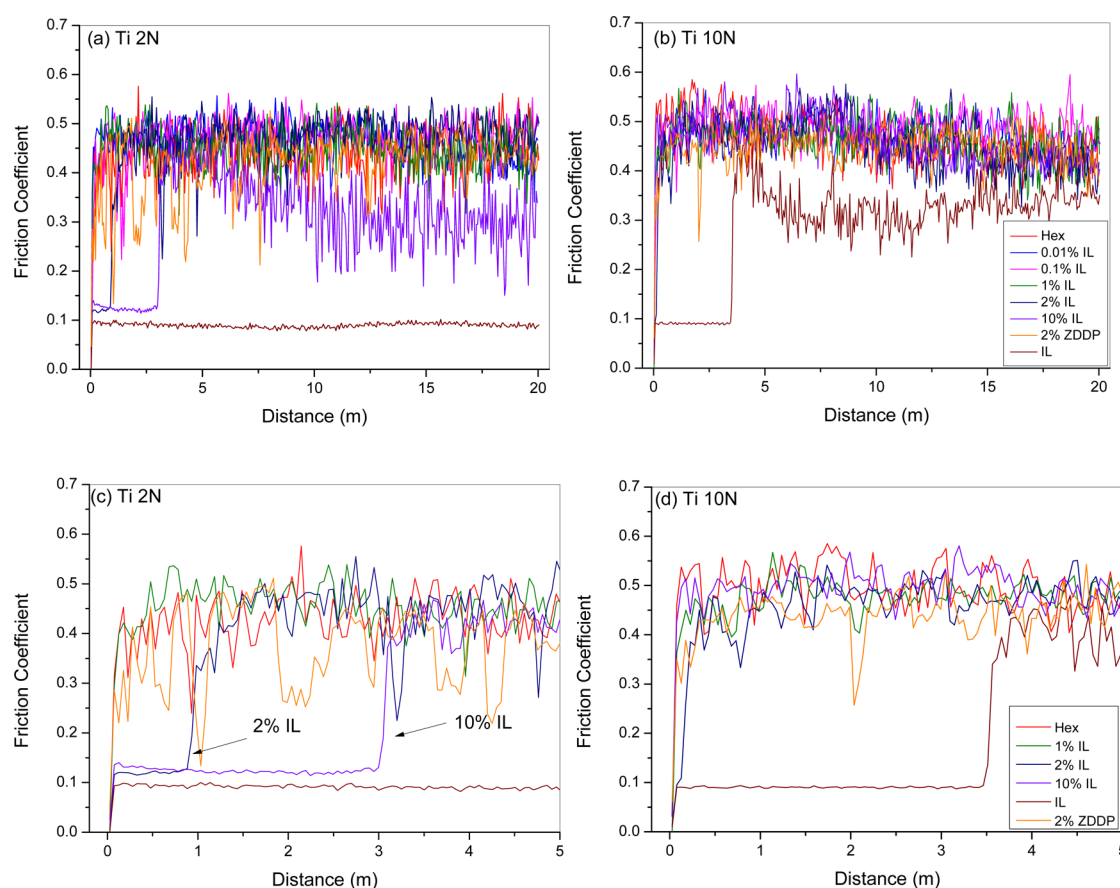
^aThe error is typically $\pm 10\%$.

Figure 3. Macroscale friction coefficient vs sliding distance for a silicon nitride ball sliding on a titanium alloy surface for 20 m with a normal load of (a) 2 and (b) 10 N. Friction coefficient vs sliding distance at different loads for the first 5 m are shown in parts c and d. The sliding speed is 50 mm/s.

For the IL–hexadecane mixtures, the normal force–separation profiles are concentration dependent. For the 0.01 mol % mixture, the normal force profile is similar to pure hexadecane. For all higher IL concentrations, a push-through layer is detected as per the pure IL, meaning that the IL ions adsorb to the titania and form a boundary layer. The force required to rupture the boundary layer increases with the IL concentration. This is consistent with the ion adsorption density increasing with the bulk IL concentration.³⁷ As the density of adsorbed ions increases so does the strength of lateral interactions between ions which results in a more robust boundary layer that remains in place up to higher forces.

Nanotribology. The nanotribology of titania lubricated by pure hexadecane, pure IL, and the IL–hexadecane mixtures was investigated by AFM friction measurements, c.f. Figure 2. For typical engines and disk brakes, surface contact stresses are generally below 0.5 GPa,⁵⁷ which corresponds to a normal force (F_N) of ~ 70 nN. As such, the normal force regime up to 90 nN meets the requirements for common applications and 300 nN provides guidance for extreme conditions. The friction coefficient, μ , is determined from the gradient of the lateral force (F_L)–normal force data in the linear regime, c.f. Table 1.

For pure hexadecane, there is no obvious change for the gradient of F_L/F_N up to 300 nN, and the friction coefficient (2.0) is very high. This reveals that, as expected, hexadecane is not a good lubricant for the titania surface. It can be pushed through even at low normal loads resulting in direct contact between the two sliding surfaces.

For the pure IL, at low normal forces, F_L is only 20% of that for pure hexadecane up to 90 nN, which means the IL lubricates titania more effectively than hexadecane over industrially relevant loads. At low loads the friction coefficient determined here for titania (0.4) is comparable to that found previously for silica (0.34),³⁸ indicating the AFM tip slides on a similar adsorbed ion layer for both surfaces. F_L increases significantly once the normal force is 160 nN and higher. This is consistent with the “push-through” event observed at similar forces in the normal force–separation profiles (Figure 1h), and confirms that the adsorbed IL boundary layer at the IL–titania interface can be displaced at high normal forces which leads to contact between the sliding surfaces and high friction.

The lateral force depends on the IL concentration in IL–hexadecane mixtures. For the 0.01 mol % mixture, the friction coefficient is constant and comparable to pure hexadecane. This

Table 2. Macroscale Friction Coefficients of IL–Hexadecane and ZDDP–Hexadecane Mixtures

	2 N		10 N	
	before break	after break	before break	after break
hexadecane	0.45 ± 0.05		0.46 ± 0.05	
0.01% IL	0.46 ± 0.04		0.47 ± 0.04	
0.1% IL	0.47 ± 0.05		0.48 ± 0.04	
1.0% IL	0.45 ± 0.05		0.46 ± 0.05	
2.0% IL	0.11 ± 0.02	0.47 ± 0.05	0.45 ± 0.06	
10% IL	0.12 ± 0.02	0.35 ± 0.07	0.47 ± 0.05	
pure IL	0.09 ± 0.01		0.09 ± 0.01	0.33 ± 0.04
2.0% ZDDP	0.42 ± 0.06		0.45 ± 0.04	

is in line with the normal force–separation profile (Figure 1) and confirms the AFM tip and titania substrate are in contact at all loads because the IL concentration is too low for a boundary layer to form.

For the 0.1 mol % IL–hexadecane mixture, when F_N is lower than 50 nN F_L is close to the pure IL data, c.f. Figure 2a. However, when the normal force is increased to more than ~50 nN, the lateral force increases rapidly and then follows the data obtained for hexadecane indicating contact between the surfaces. This is consistent with the “push-through” event at the same normal force in the normal force–separation profiles (Figure 1c).

The form of the data is similar at higher IL concentrations, but the normal load at which the boundary layer is ruptured increases with IL concentration. The forces at which the breakpoints in the friction data occur are consistent with the push-throughs at 80–120 nN in the normal force–separation profiles (Figure 1d–f). At low IL concentrations, the adsorbed ion density is low, meaning the boundary layer is fragile and is thus ruptured at low forces. At higher IL concentrations the surface ion adsorption density must be higher, resulting in a better formed boundary layer and stronger lateral interactions between ions. This means the resistance of the adsorbed ions to collective displacement is greater. In fact, at titania when the IL concentration is 1 mol % and above, the surface excess is high enough to form a robust boundary layer that remains in place up to 90 nN, which is comparable to the contact stresses of common applications. Although the boundary layers are not as ordered as the pure IL which leads to higher friction coefficients (c.f. Table 1), these IL–hexadecane mixtures are more cost-effective, and thus have wider industrial appeal, especially in the low-end markets.

In previous studies on a silica surface for the same pure IL and IL–hexadecane mixtures of 2 mol % and above, no push-through events were found up to a normal load of (at least) 300 nN.³⁷ However, the nanotribology results at titania here suggest that the adsorbed IL boundary layer can only remain at the interface up to the critical normal load, even for the pure IL. This is attributed to the lower charge density of titania compared to silica leading to reduced strength of ion adsorption and thus weaker boundary layers.

Macrotribology. The macrotribology of titanium alloys lubricated by pure hexadecane, pure IL, IL–hexadecane, and 2.0 mol % ZDDP–hexadecane mixtures was tested using a pin-on-disk tribometer. The sliding speed was 50 mm/s, and the applied normal loads were 2 and 10 N. Given the difficulties of monitoring wear of the AFM tip used in nanotribology and the ball used in macrotribology over the course of an experiment, the initial contact stresses were matched for the nano- and macrotribology studies. The 2 N load at the macroscale

corresponds to a Hertzian contact stress of 0.68 GPa, which is approximately equal to a load of 90 nN at nanoscale in Figure 2. A 10 N load at the macroscale corresponds to a Hertzian contact stress of 1.2 GPa and is comparable with the 160 nN load at the nanoscale.

The macroscale friction coefficient versus sliding distance for the systems tested is shown in Figure 3. Comparison of the macroscale friction coefficients in Table 2 with the nanoscale friction coefficients in Table 1 reveals that macroscale friction coefficients are about 2–4 times smaller. As the surface roughness for titanium alloy surfaces used at the macroscale is much higher than the thickness of IL layers estimated using the Hamrock and Dowson model (c.f. Supporting Information (SI)), all hexadecane, IL–hexadecane, and the pure IL are in the boundary lubrication regime at both the macro- and nanoscales. The differences found in friction coefficients at both scale lengths are thus likely because the sliding speed at the macroscale is 3 orders of magnitude higher than that at the nanoscale, which leads to lower friction coefficients according to the generalized Stribeck curve.^{58,59}

Figure 3a reveals that at 2 N the macroscale friction coefficient for hexadecane is 0.45. This is comparable with previous work⁶⁰ and is more than four times higher than the pure IL friction coefficient 0.09. The lower friction coefficient for the pure IL is consistent with nanotribology results under the same contact stress, indicating that the IL boundary layer adsorbs more strongly at the interface than pure hexadecane. While the pure IL nanoscale friction coefficients are similar for titania and silica,³⁸ the macroscale friction coefficient for titania is about one-third higher than silica's. This is because macrotribology is significantly influenced by the substrate properties, such as the surface roughness, whereas nanofriction is mainly affected by the boundary layer structure determined by the liquid properties. The roughness of titanium alloys is more than one hundred times higher than that of silicon wafers, which produces higher macroscale friction coefficients.

The friction coefficients for 1 mol % and below IL–hexadecane mixtures are the same as pure hexadecane, i.e. the added IL has no measurable effect. This indicates that the IL boundary layers formed at these concentrations are too weak to prevent surface contact. This result is in line with the nanoscale results shown in Figures 1 and 2, where at the same contact stress (90 nN) the boundary layer is ruptured at low force and the friction coefficients increase significantly due to the direct contact of the surfaces.

At 2 N, when the IL concentration in hexadecane is increased to 2 and 10 mol %, the friction coefficients are initially similar to that of the pure IL but increase rapidly at 1 and 3 m respectively, and are then comparable with the pure hexadecane value. Thus, at these concentrations the boundary layer remains

in place for short distances, but is then ruptured. The distance at which the rapid increase occurred varied slightly for different experiments. These results are consistent with the boundary layer displacement observed for the same IL–hexadecane mixtures in the nanoscale tests under similar contact stresses (Figures 1e, f and 2); only the displacement of the boundary layer observed at the macroscale occurs at a slightly lower contact stress, likely due to the increased sliding distance and surface roughness. The rapid increase in friction coefficient suggests that the boundary layer is ruptured suddenly and cannot reform.

At 10 N, the friction coefficient for pure hexadecane has no significant change, but for all the IL–hexadecane mixtures, the friction coefficients are the same as pure hexadecane, i.e. the added IL has no measurable effect. This is confirmed by the nanoscale tests. As the contact stress of 10 N at the macroscale is equal to 160 nN at the nanoscale, the adsorbed IL boundary layers have been squeezed out at this contact stress as shown in Figures 1 and 2, leading to surface contact. For the pure IL, at this load a sudden increase in friction coefficient is observed at ~ 3.5 m, consistent with the displacement of IL boundary layer observed under the similar contact stress at the nanoscale, c.f. Figures 1h and 2.

The macrotribology of 2 mol % ZDDP–hexadecane mixture was also investigated as shown in Figure 3. At both low and high loads, the friction coefficients are constantly high and comparable to pure hexadecane. This means at the same concentrations, the IL works better than ZDDP as a lubricant additive to reduce friction, at least for low normal loads and short sliding distances.

Note that the fluctuations of the macroscale friction coefficients shown in Figures 3 and Table 2 are relatively large, except for the pure IL and high IL concentration mixtures before break. This is likely caused by the formation and periodic, localized fracture of a transfer layer from the titanium alloy disk to the silicon nitride ball when they are rubbing against each other, as observed previously.^{42,60,61} Much smaller fluctuations of the macroscale friction coefficients are found for the pure IL and IL–hexadecane mixtures before the push-through of the IL boundary layers. This also suggests that the IL boundary layer can protect the surfaces from direct contact and thus prevent the formation of the transfer layer.

Macroscale Wear. The wear tracks on the titanium alloy surfaces after the macrotribology tests were measured using 3D optical profiler and SEM. The morphologies of the wear tracks are shown in Figure S2, SI, which show the widths of the wear tracks at different conditions at 10 N load are similar, all ~ 500 μm . The wear depths at both 2 and 10 N are shown in Figure 4. Data is not shown for the pure IL at 2 N as no wear was apparent, consistent with previous observations for silica,³⁸ meaning the adsorbed boundary layer in the pure IL protects the titania surface at this load.

The wear depth is greater at 10 N than at 2 N for all systems. Compared to dry contact, the presence of hexadecane reduces the wear depth by only $\sim 10\%$ and $\sim 15\%$ at 2 and 10 N, respectively. Generally, the IL–hexadecane mixtures show better antiwear performance than pure hexadecane, most notably at higher IL contents.

At 2 N, the errors associated with the wear depths are high due to their small values. However, there appears to be a general trend of wear depth decreasing with IL concentration, as may be expected based on the results above. When the normal load is increased to 10 N, the wear depth decreases with

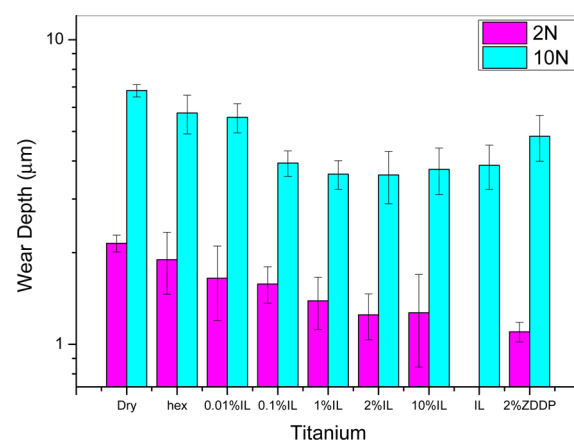


Figure 4. Wear depth for the IL–hexadecane mixtures at titanium alloy surfaces.

IL concentration until a concentration of ~ 1 mol % is reached, then plateaus. As both the nanoscale and macroscale friction measurements show that the surfaces are in direct contact at this normal load, the reduction in wear with IL concentration means the IL acts as a lubricant in the traditional sense. Most likely, at 1 mol % the IL concentration is sufficiently high that it can adsorb to small debris in a surfactant like fashion and remove them from the contact zone, thereby reducing wear. In this scenario, it is reasonable to expect a corresponding trend at the lower (2 N) load; when the error bars are taken into account it is possible that this trend holds. Similar effects have been noted previously at the same load at silica surfaces.³⁸ It is also possible that an effective tribolayer is formed when the IL concentration is 1 mol % and above and thus resists further wear, which has been observed in several phosphonium-based IL and IL–oil mixture macrotribology studies.^{8,28,30–32} However, as both the cation and anion of the investigated IL are based on the phosphonium ion, it is difficult to determine whether decomposition of the cation, anion, or both is responsible for the layer.

For the 2 mol % ZDDP–hexadecane mixture, at 2 N the wear depth is very low, likely due to the formation of a tribolayer.⁹ However, at 10 N, the wear depth increases significantly, and is 33% higher than that of the 2 mol % IL–hexadecane mixture, indicating that any tribolayer is thin and readily displaced at high loads.⁴ Together, the wear results show that the IL is an effective antiwear additive for titanium alloys and has better antiwear performance than ZDDP mixtures of the same concentration at high loads as the IL adsorbed layers are better able to protect the surface than the reacted ZDDP tribolayer.

CONCLUSIONS

Nanotribology has been combined with macrotribology to demonstrate that an IL boundary layer adsorbed from IL–oil mixtures effectively lubricates titania at contact stresses comparable to common industrial applications. At both nano- and macroscales, pure hexadecane is squeezed out at all contact stresses, leading to the direct surface contact and high friction. The pure IL lubricates the titania surface effectively at low contact stresses, but the friction increases markedly once the contact stress is higher than 1.2 GPa due to the adsorbed IL layer being displaced. For IL–hexadecane mixtures, the critical contact stress is lower than the pure IL and increases with IL

concentration. The performance of ZDDP as lubricant additives was investigated under the same macrotribological conditions and revealed that, at the same concentration, the IL is a better lubricant additive for titania surfaces than ZDDP, especially at high contact stresses. All these results suggest that IL ions interact more strongly with titania than conventional lubricant oil molecules; therefore, the formed IL boundary layers can protect the titania surface effectively at contact stresses required for common applications. The outcomes of this study pave the way for the commercialization of IL lubricants and application of titanium and alloys as moving components in industry. However, the IL boundary layer can be displaced at high contact stresses, as such new oil miscible ILs, e.g. cations/anions with shorter alkyl chain and thus stronger interactions with titania, need to be designed for applications in extreme conditions.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acssuschemeng.6b01383](https://doi.org/10.1021/acssuschemeng.6b01383).

Figure S1: AFM image of titania coated silicon wafer. Calculation of Herzian contact stress. Hamrock and Dowson model. Figure S2: SEM images of the contact area of titanium alloy in air (a) and lubricated with hexadecane (b), 2% IL (c), and 2% ZDDP (d) at a normal load of 10 N (PDF)

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Notes

The authors declare no competing financial interest.

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